

Kevin Cimorelli
Site Vice President

Susquehanna Nuclear, LLC
769 Salem Boulevard
Berwick, PA 18603
Tel. 570.542.3795 Fax 570.542.1504
Kevin.Cimorelli@TalenEnergy.com



July 1, 2021

Attn: Document Control Desk
U. S. Nuclear Regulatory Commission
Washington, DC 20555-0001

10 CFR 50.73

**SUSQUEHANNA STEAM ELECTRIC STATION
LICENSEE EVENT REPORT 50-388/2017-010-02
UNIT 2 LICENSE NO. NPF-22
PLA-7951**

Docket No. 50-388

Attached is the second supplement to Licensee Event Report (LER) 50-388/2017-010 that reported an event involving drifting of Reactor Pressure Steam Dome – Low pressure switches (Microswitch 2). The condition was reported as a condition prohibited by Technical Specifications in accordance with 10 CFR 50.73(a)(2)(i)(B), a condition that could have prevented fulfillment of a safety function in accordance with 10 CFR 50.73(a)(2)(v)(D), a common cause inoperability of independent trains or channels in accordance with 10 CFR 50.73(a)(2)(vii), and a single cause that could have prevented fulfillment of safety functions of trains or channels in different systems in accordance with 10 CFR 50.73(a)(2)(ix)(A). The reason for this supplement is a revised causal analysis that resulted in a modification to the cause and associated corrective actions.

There were no actual consequences to the health and safety of the public as a result of this event.

This letter contains no new or revised regulatory commitments.

A handwritten signature in black ink, appearing to read "K. Cimorelli", written over a horizontal line.

K. Cimorelli

Attachment: LER 50-388/2017-010-02

Copy: NRC Region I
Mr. C. Highley, NRC Senior Resident Inspector
Ms. A Klett, NRC Project Manager
Mr. M. Shields, PA DEP/BRP

**LICENSEE EVENT REPORT (LER)**

(See Page 3 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form <https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk all: omb_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Susquehanna Steam Electric Station 2

2. Docket Number

05000388

3. Page

1 of 3

4. Title Condition Prohibited by Technical Specifications Due to Drift of Reactor Pressure Switches

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
12	05	2017	2017	010	02	07	01	2021	Facility Name	Docket Number
										05000
										05000

9. Operating Mode

1

10. Power Level

100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☒ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☒ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☒ 50.73(a)(2)(ix)(A)	

☐ **Other** (Specify here, in Abstract, or in NRC 366A).**12. Licensee Contact for this LER****Licensee Contact**

C. E. Manges, Jr., Principle Engineer – Nuclear Regulatory Affairs

Phone Number (Include Area Code)

(570) 542-3089

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	JE	PS	GE/Cameron	Y					

14. Supplemental Report Expected☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)**15. Expected Submission Date**

Month	Day	Year

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On December 5, 2017, the Unit 2 "C" and Unit 2 "D" Reactor Steam Dome Pressure – Low permissive pressure switches, Microswitch 2 (SW2), were found outside of the Technical Specification (TS) 3.3.5.1 allowable value. Both drifted outside of the upper allowable value which is intended to ensure that the reactor dome pressure has fallen to a value below the Core Spray and Residual Heat Removal (RHR)/Low Pressure Coolant Injection (LPCI) maximum design pressures to preclude over-pressurization of the low pressure systems prior to low pressure injection initiation.

Based on the information available, the condition existed for longer than allowed by TS 3.3.5.1 and TS 3.5.1. As such, this is a condition prohibited by TS and is being reported in accordance with 10 CFR 50.73(a)(2)(i)(B). This is also considered a common cause inoperability of independent trains or channels (10 CFR 50.73(a)(2)(vii)), a condition that could have prevented fulfillment of a safety function (10 CFR 50.73(a)(2)(v)(D)), and a single cause that could have prevented fulfillment of safety functions of trains or channels in different systems (10 CFR 50.73(a)(2)(ix)(A)).

The cause was determined to be instrument setpoint drift due to temperature/humidity changes and mechanical hysteresis as a result of the original design inputs not accounting for temperature and humidity effects on the switch. Interim compensatory actions include performing calibration using the quarterly calibration procedures every 45 days on Unit 1 and every 30 days on Unit 2. Final corrective action will revise the allowable value for TS 3.3.5.1, Functions 1.c, 1.d, 2.c, and 2.d.

There were no actual consequences to the health and safety of the public as a result of this event.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Susquehanna Steam Electric Station, Unit 2	05000-388	2017	010	02

NARRATIVE**CONDITIONS PRIOR TO EVENT**

Unit 1 – Mode 1, approximately 100 percent Rated Thermal Power

Unit 2 – Mode 1, approximately 100 percent Rated Thermal Power

There were no structures, systems, or components that were inoperable at the start of the event that contributed to the event.

EVENT DESCRIPTION

Susquehanna Steam Electric Station (SSES) had been utilizing International Telephone and Telegraph (ITT)-Barton 288A pressure switches in the Reactor Steam Dome Pressure – Low channels that provide the injection permissive for the Core Spray system [EIS System Identifier: BM] (Technical Specification (TS) 3.3.5.1, Function 1d) and the Residual Heat Removal (RHR)/Low Pressure Coolant Injection system (LPCI) [EIS System Identifier: BO] (TS 3.3.5.1, Function 2d). Each pressure switch has two microswitches, low (SW1) and high (SW2). All eight obsolete ITT-Barton 288A pressure switches were replaced with General Electric (GE) recommended Cameron-Barton 288A pressure switches between September 6, 2017 and November 15, 2017. The Cameron Barton 288A switches were bench tested prior to installation and calibration checked at the time of installation. Subsequent calibration checks were performed at intervals more frequent than the quarterly TS required calibrations. During these subsequent calibration checks, SW2 of the Unit 2 “C” (PIS-B21-2N021C) and Unit 2 “D” (PIS-B21-2N021D) pressure switches [EIS System/Component Identifier: JE/PS], were found outside of the TS allowable value. A timeline of events is as follows:

September 20, 2017 - the obsolete ITT-Barton 288A pressure switch for PIS-B21-2N021C was replaced with a GE recommended Cameron-Barton 288A pressure switch.

October 9, 2017 - the obsolete ITT-Barton 288A pressure switch for PIS-B21-2N021D was replaced with a GE recommended Cameron-Barton 288A pressure switch.

October 20, 2017 - PIS-B21-2N021C and PIS-B21-2N021D were calibration checked and were found within the TS allowable value.

December 5, 2017 - PIS-B21-2N021C and PIS-B21-2N021D were found outside of the TS allowable value during the calibration check. Both drifted outside of the upper allowable value which is intended to ensure that the reactor dome pressure has fallen to a value below the Core Spray and RHR/LPCI maximum design pressures to preclude over-pressurization of the low pressure systems prior to low pressure injection initiation. PIS-B21-2N021C exceeded the TS allowable value by 1.3 psi. PIS-B21-2N021D exceeded the TS allowable value by 2.8 psi. Both switches were adjusted to within the TS allowable value.

After submitting LER 50-388/2017-010-00, two additional instances of drift outside of the TS allowable value were identified. Specifically, the Unit 1 “B” (PIS-B21-1N021B) switch was found outside the TS allowable value on June 5, 2018 and the Unit 2 “C” (PIS-B21-2N021C) switch was again found outside of the TS allowable value on June 6, 2018. These conditions are addressed in LER 50-387(388)/2018-005.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Susquehanna Steam Electric Station, Unit 2	05000-388	2017	010	02

NARRATIVE

Based on the information available, the condition existed for longer than allowed by TS 3.3.5.1 and TS 3.5.1. As such, this is a condition prohibited by TS and is being reported in accordance with 10 CFR 50.73(a)(2)(i)(B). In addition, since two Unit 2 channels were outside of the TS allowable values and impacted both Core Spray and LPCI functions, this is also considered a common cause inoperability of independent trains or channels (10 CFR 50.73(a)(2)(vii)), a condition that could have prevented fulfillment of a safety function (10 CFR 50.73(a)(2)(v)(D)), and a single cause that could have prevented fulfillment of safety functions of trains or channels in different systems (10 CFR 50.73(a)(2)(ix)(A)).

CAUSE OF EVENT

The direct cause was determined to be instrument setpoint drift due to temperature/humidity changes and mechanical hysteresis. The apparent cause involved the original design inputs not accounting for temperature and humidity effects on the switch.

ANALYSIS/SAFETY SIGNIFICANCE

An engineering evaluation concluded that all components in the RHR/LPCI and Core Spray would have been able to withstand a pressure of 457.8 psig, which was the maximum recorded pressure for the instrument drift. Therefore, Core Spray and RHR/LPCI would have been able to perform their safety function. This event will not be counted as a safety system functional failure for the NRC performance indicator based on the engineering analysis that shows there was no loss of ability to fulfill the safety function.

CORRECTIVE ACTIONS

Interim compensatory actions include performing calibration using the quarterly calibration procedures every 45 days on Unit 1 and every 30 days on Unit 2.

Final corrective action will revise the allowable value for TS 3.3.5.1, Functions 1.c, 1.d, 2.c, and 2.d.

COMPONENT FAILURE INFORMATION

The switches that drifted are Cameron-Barton 288A pressure indicating switches manufactured by Cameron.

PREVIOUS SIMILAR EVENTS

LER 50-388(387)/2015-001-01, "Condition Prohibited by Technical Specifications Due to Drift of Reactor Pressure Steam Dome-Low Switches", dated February 10, 2016.

LER 50-387(388)/2018-005-01, "Condition Prohibited by Technical Specifications Due to Drift of Reactor Pressure Switches", dated December 16, 2019.